

TIC 399544063

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 0.692904 d
Epoch (t0) 2882.3628 BTJD
Duration 0.0400 d
Depth 3.130 % (31,295 ppm)
SNR 947.4

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

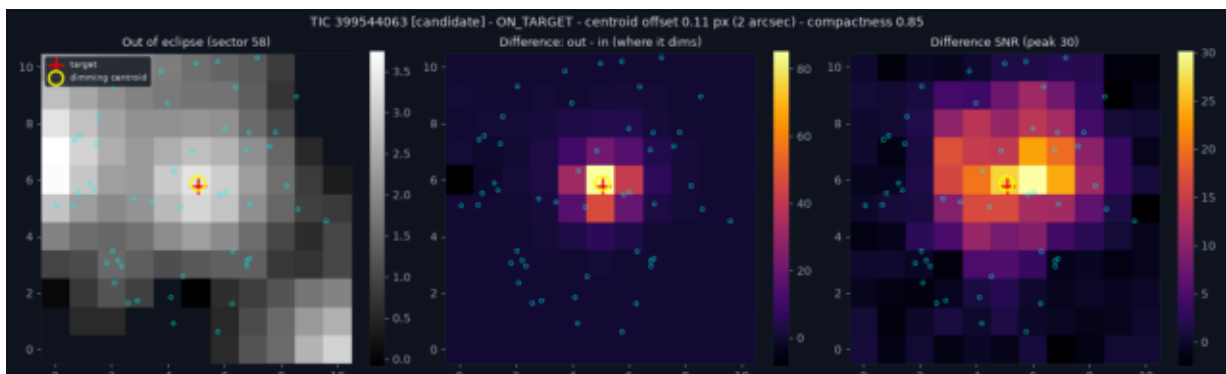
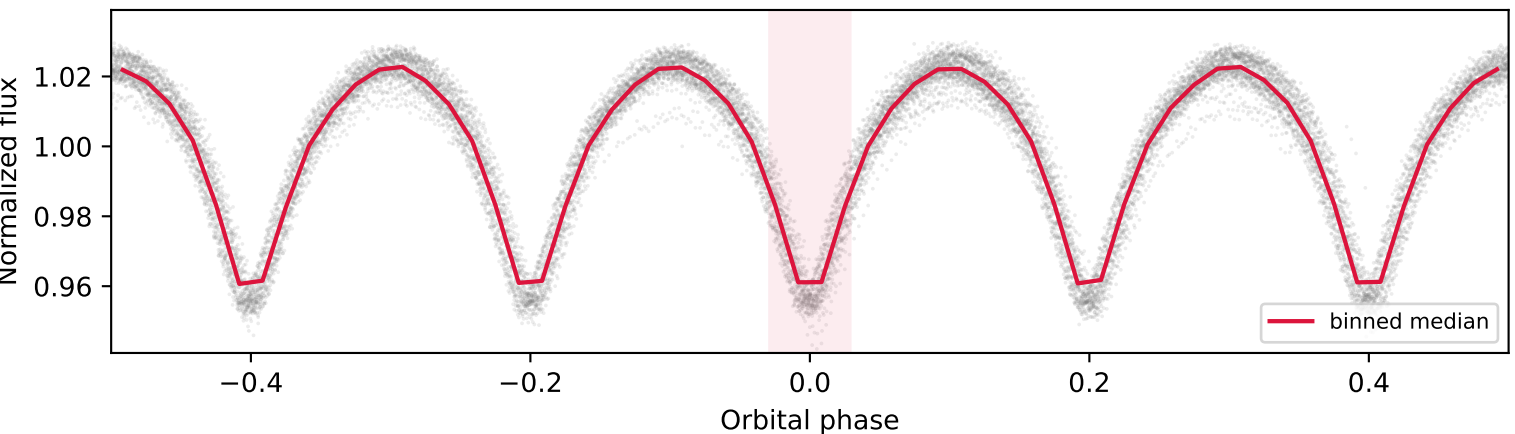
Eclipses observed 66 (gate ≥ 3)
Robust depth ratio 1.26 (gate ≥ 0.50)
Robust depth sigma 58.0 (gate ≥ 7.0)
Odd/even z-score 1.82 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 58 light-curve dip in this sector: 36.7 sigma t0 refined by -0.3975 phase
Difference-image peak SNR 30.25 (gate ≥ 5.0) Compactness 0.85 (gate ≥ 0.50)
Dimming centroid offset 0.11 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: none within the cutout

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.